

COLLEGE MINOR PROJECT

Weather Data Analysis and Prediction

Using Machine Learning to Forecast Temperature, Rainfall and Weather Trends from Historical Data

Submitted by: Your Name

Course: B.Tech / B.Sc (CS) — 2nd Year

Guided by: Faculty Name, Department of Computer Science

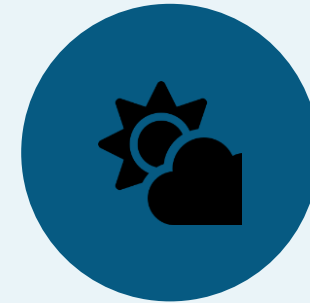
Your College / University Name



What Is Weather Data Analysis?

Weather data analysis is the process of examining historical meteorological records — temperature, humidity, wind speed, pressure and rainfall — to understand patterns and relationships in the atmosphere. When combined with machine learning, this analysis can be used to predict future weather conditions.

- Weather affects agriculture, transport, aviation, and daily human activity.
- Traditional forecasting relies on complex physical simulations that are costly to run.
- Data-driven models learn directly from years of recorded weather data.



Data-Driven Forecasting

Turning raw sensor & station readings into actionable weather predictions.

Why Traditional Forecasting Falls Short



Conventional Numerical Weather Prediction (NWP) systems require massive computing power, expert calibration, and still struggle with short-term, localized accuracy. There is a need for a lightweight, data-driven system that can learn patterns directly from historical records and generate reliable, low-cost predictions for a specific region.

High

Computational cost of physics-based models

Low

Accuracy for hyper-local, short-term forecasts

Slow

Turnaround for updating predictions in real time

What This Project Aims to Achieve

1



Analyze Historical Data

Explore multi-year weather records to identify seasonal trends and correlations.

2



Clean & Prepare Data

Handle missing values, outliers and normalize features for modelling.

3



Build Prediction Models

Train and compare ML models to forecast temperature and rainfall.

4



Evaluate & Visualize

Measure model accuracy and present results through clear visualizations.

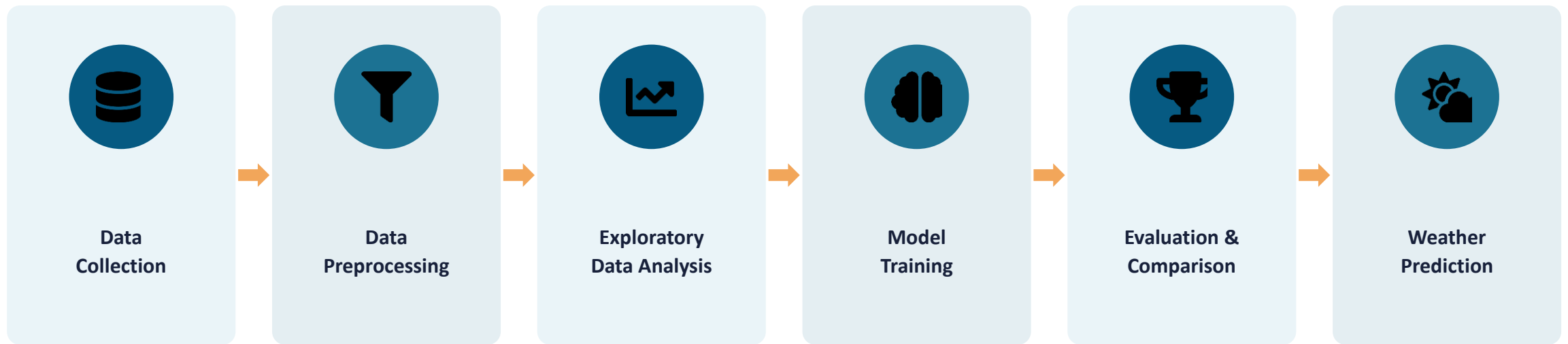
Existing Approaches to Weather Forecasting



Weather prediction methods have evolved from purely physical simulations to hybrid, data-driven systems:

Approach	How It Works	Limitation
Numerical Weather Prediction (NWP)	Solves physical equations of atmospheric dynamics on supercomputers	Extremely resource-intensive; coarse spatial resolution
Statistical Methods (ARIMA, Regression)	Fits mathematical trends to historical time-series data	Struggles with non-linear, complex weather patterns
Classical Machine Learning (RF, SVM)	Learns patterns from labelled historical features	Needs good feature engineering; limited long-range memory
Deep Learning (LSTM, CNN)	Learns temporal/spatial patterns automatically from raw data	Requires large datasets and higher training time

End-to-End Workflow of the Project



A structured pipeline takes raw meteorological data through cleaning and analysis before training multiple models, comparing their performance, and generating final weather predictions.

The Technology Stack



Python

Core programming language for the entire pipeline



Pandas & NumPy

Data loading, cleaning and numerical computation



Matplotlib & Seaborn

Visualizing trends, distributions and correlations



Scikit-learn / TensorFlow

Building and training prediction models



Jupyter Notebook

Interactive development and experimentation



**Kaggle / OpenWeatherMap
API**

Source of historical & live weather data

Understanding the Data

The project uses a historical daily weather dataset (e.g. Kaggle's Weather Dataset or a national meteorological department archive) covering several years of station readings.

Feature	Description
Date	Day of observation used to derive seasonal/trend features
Temperature (Min/Max)	Daily minimum and maximum temperature in °C
Humidity	Relative humidity percentage recorded at the station
Wind Speed & Direction	Speed (km/h) and direction of surface wind
Atmospheric Pressure	Sea-level pressure in hPa
Rainfall	Precipitation recorded in millimetres (target for rain prediction)



5+ Years

of daily historical records

10+ Features

describing daily atmospheric state

2 Targets

temperature & rainfall prediction

Preparing Raw Data for Modelling

1 Handle Missing Values

Impute or drop records with missing temperature, humidity or rainfall readings.

2 Remove Outliers

Detect and treat sensor errors and extreme anomalies using statistical thresholds.

3 Feature Engineering

Derive new features such as month, season, and rolling averages from the date.

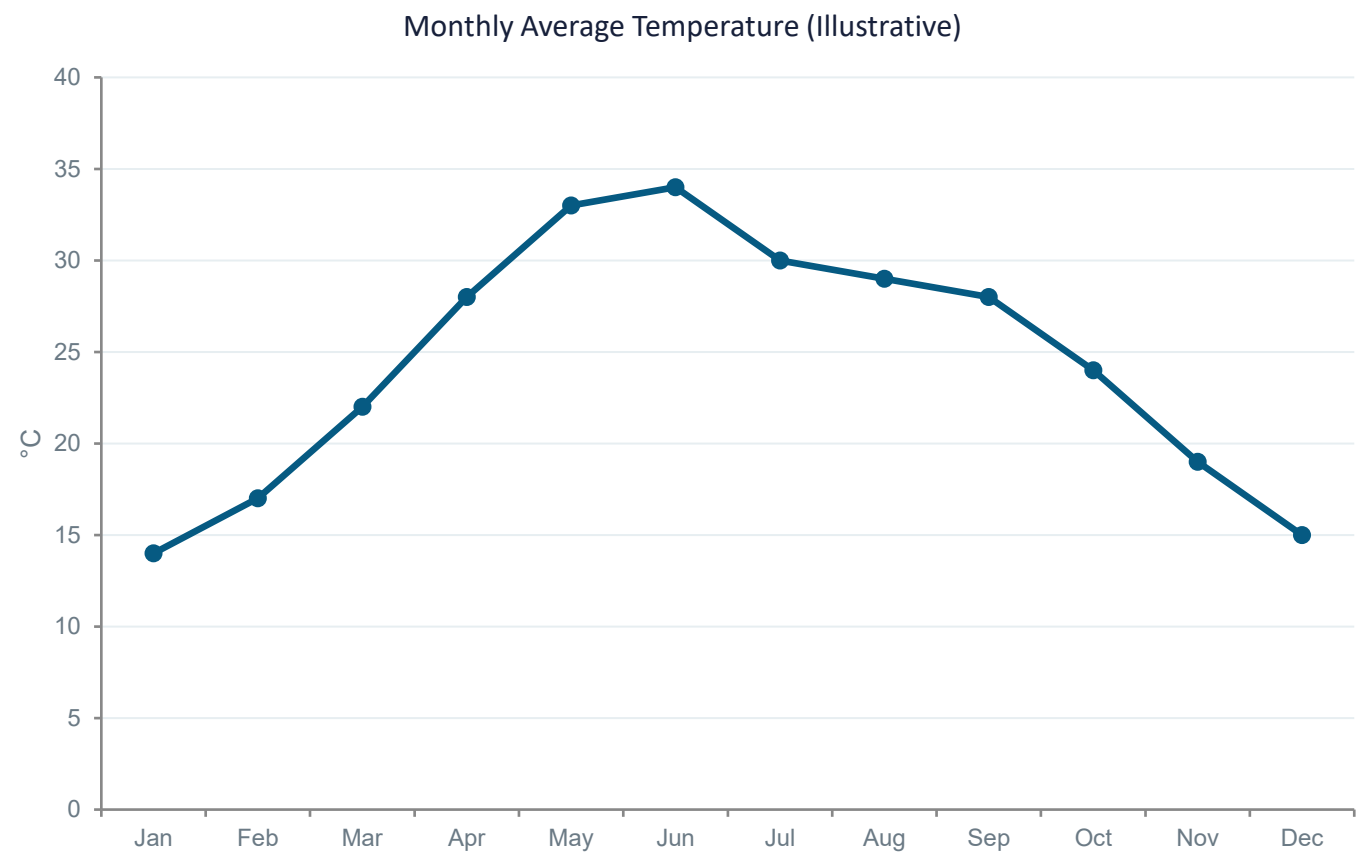
4 Normalize / Scale Features

Bring all numerical features to a comparable range for stable model training.

5 Train-Test Split

Split data chronologically (e.g. 80:20) to fairly evaluate prediction performance.

Uncovering Seasonal Temperature Trends



Key Insights

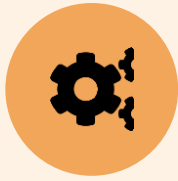
- Temperature peaks in June, aligning with the pre-monsoon period.
- Rainfall is heavily concentrated in the June–September monsoon window.
- Winter months (Dec–Feb) show the lowest variance in daily readings.
- Humidity strongly correlates with rainfall onset, a useful predictive feature.

Algorithms Used for Prediction



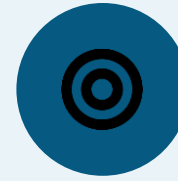
Linear Regression

Baseline model to capture straightforward linear relationships between features and temperature.



Decision Tree / Random Forest

Ensemble of trees that captures non-linear interactions between humidity, pressure and rainfall.



K-Nearest Neighbours

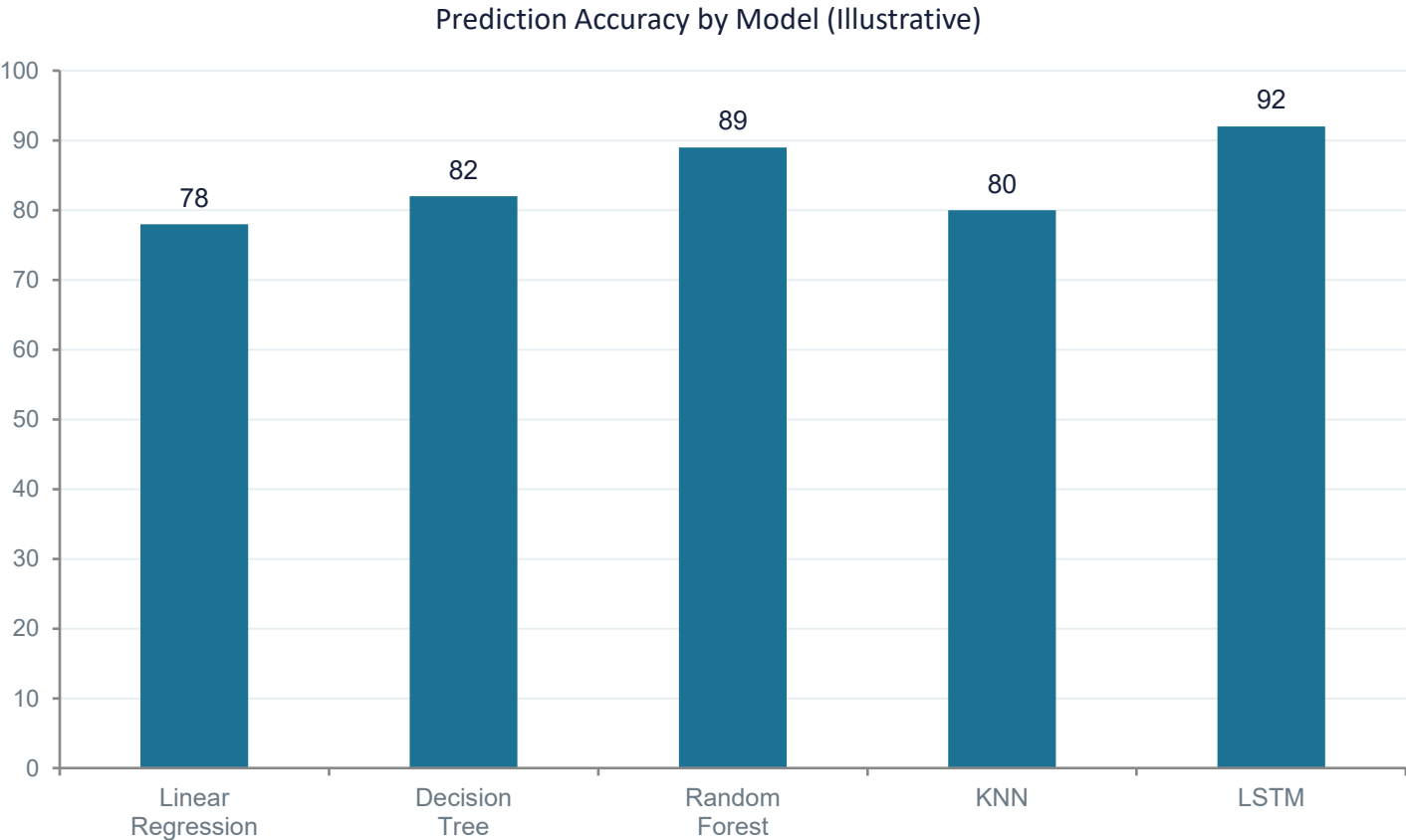
Predicts weather based on the most similar historical days in the feature space.



LSTM (Deep Learning)

Recurrent neural network that learns long-term temporal dependencies in sequential weather data.

Model Performance Comparison



Best Performer

LSTM — 92%

The LSTM network achieved the highest accuracy by learning temporal dependencies across consecutive days, closely followed by Random Forest, which handled non-linear feature interactions well. Evaluation used RMSE and R^2 alongside accuracy for a complete picture.

Where This System Adds Value



Agriculture

Helping farmers plan sowing and irrigation around predicted rainfall.



Aviation & Transport

Supporting flight and route planning with short-term forecasts.



Disaster Management

Early warnings for heavy rainfall, storms and heatwaves.



Daily Life & Energy

Assisting personal planning and renewable energy load forecasting.

Summary and Future Scope



Conclusion

- A complete pipeline was built from raw data to trained ML models.
- LSTM and Random Forest outperformed simpler baseline models.
- The system demonstrates that data-driven forecasting is a viable, low-cost complement to traditional methods.



Future Scope

- Integrate live data via weather APIs for real-time predictions.
- Extend the model to forecast extreme events like storms and heatwaves.
- Deploy as a web or mobile app for public and farmer-facing use.

Thank You

Questions & Discussion Welcome

REFERENCES

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